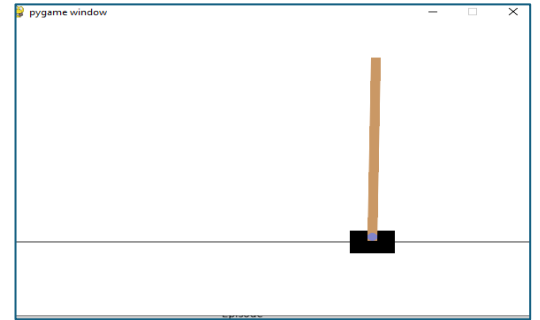


CartPole_Assignment

Introduction

The CartPole problem is a classic reinforcement learning environment. The objective is to keep a pole balanced upright on a moving cart for as long as possible. The cart can be pushed left or right, and the environment provides a reward for every time step the pole remains upright. If the pole falls past a certain angle or the cart moves too far from the center, the episode terminates.



Markov Decision Process (MDP) Formulation

To solve this problem using reinforcement learning, we model it as an MDP defined by a tuple (S, A, P, R, γ) :

Algorithm: Deep Q-Network (DQN)

For this assignment, we implemented the Deep Q-Network (DQN) algorithm. DQN extends standard Q-learning by using a neural network to approximate the Q-value function $Q(s, a)$.

➤ Network architecture:

The Q-network is a Multi-Layer Perceptron (MLP) built with PyTorch. It takes the 4-dimensional state vector as input and outputs the estimated Q-values for the 2 possible actions. The architecture consists of:

- Input Layer (4 nodes) → Hidden Layer 1 (128 nodes, ReLU activation)
- Hidden Layer 1 → Hidden Layer 2 (128 nodes, ReLU activation)
- Hidden Layer 2 → Output Layer (2 nodes, Linear activation)

➤ Key points:

To stabilize training, DQN utilizes two key mechanisms:

1. **Experience Replay Buffer:** Instead of training on consecutive frames (which are highly correlated), the agent stores transitions (s, a, r, s') in a memory buffer. During training, random mini batches of size 128 are sampled, breaking the correlation between experiences and stabilizing the network.
2. **Target Network:** Two identical networks are used: a Policy Network (updated at every step) and a **Target Network** (used to calculate the target Q-value). The Target Network is updated slowly using a "soft update" approach ($\tau = 0.005$) to prevent the target values from oscillating erratically.

➤ Exploration strategy:

The agent explores the environment using an ϵ -greedy strategy. With probability ϵ , it takes a random action, and with probability $1 - \epsilon$, it exploits the known Q-values. ϵ starts at 0.9 and decays exponentially to a minimum of 0.05.

➤ Result:

The agent was trained for up to 500 episodes using an AdamW optimizer with a learning rate of 1×10^{-4} . cartpole

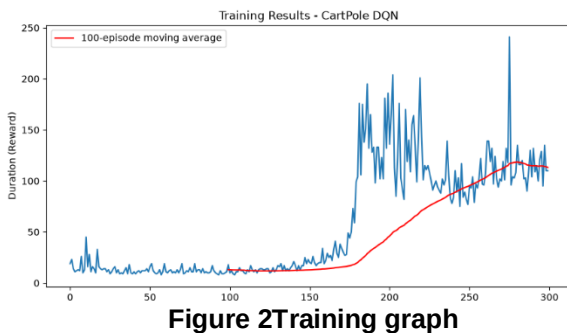


Figure 2 Training graph

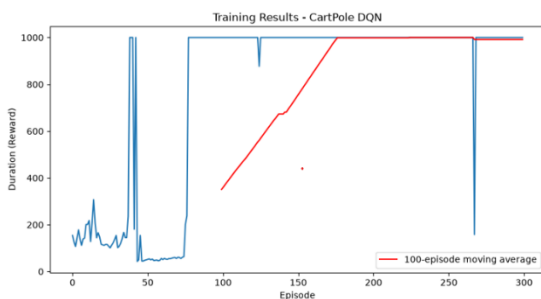


Figure 5 Training graph 2

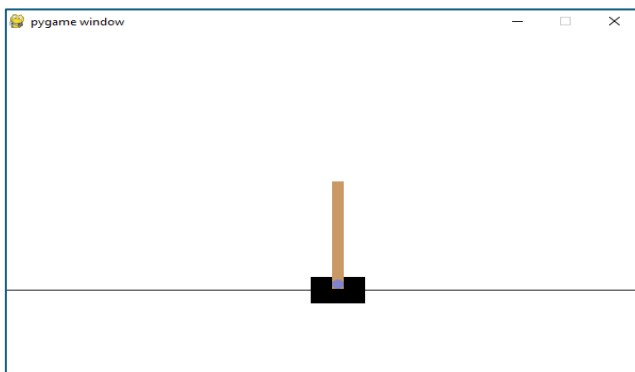


Figure 7 cartpole

```
Starting Evaluation...
Episode 1 finished with a reward of 106.0
Episode 2 finished with a reward of 117.0
Episode 3 finished with a reward of 129.0
Episode 4 finished with a reward of 110.0
Episode 5 finished with a reward of 110.0
Episode 6 finished with a reward of 117.0
Episode 7 finished with a reward of 114.0
Episode 8 finished with a reward of 99.0
Episode 9 finished with a reward of 97.0
Episode 10 finished with a reward of 111.0
Episode 11 finished with a reward of 104.0
Episode 12 finished with a reward of 104.0
Episode 13 finished with a reward of 123.0
Episode 14 finished with a reward of 123.0
Episode 15 finished with a reward of 137.0
```

Figure 1 Testing (animation)

```
Starting Evaluation...
Episode 1 finished with a reward of 500.0
Episode 2 finished with a reward of 500.0
Episode 3 finished with a reward of 500.0
Episode 4 finished with a reward of 500.0
Episode 5 finished with a reward of 500.0
Episode 6 finished with a reward of 500.0
Episode 7 finished with a reward of 500.0
Episode 8 finished with a reward of 500.0
Episode 9 finished with a reward of 500.0
Episode 10 finished with a reward of 500.0
Episode 11 finished with a reward of 500.0
Episode 12 finished with a reward of 500.0
Episode 13 finished with a reward of 500.0
Episode 14 finished with a reward of 500.0
Episode 15 finished with a reward of 500.0
```

Figure 6 testing (animation)

```
Starting Training...
Episode 10/300 - Duration (Reward): 13
Episode 20/300 - Duration (Reward): 14
Episode 30/300 - Duration (Reward): 10
Episode 40/300 - Duration (Reward): 9
Episode 50/300 - Duration (Reward): 11
Episode 60/300 - Duration (Reward): 19
Episode 70/300 - Duration (Reward): 13
Episode 80/300 - Duration (Reward): 11
Episode 90/300 - Duration (Reward): 17
Episode 100/300 - Duration (Reward): 18
Episode 110/300 - Duration (Reward): 10
Episode 120/300 - Duration (Reward): 14
Episode 130/300 - Duration (Reward): 13
Episode 140/300 - Duration (Reward): 17
Episode 150/300 - Duration (Reward): 23
Episode 160/300 - Duration (Reward): 19
Episode 170/300 - Duration (Reward): 38
Episode 180/300 - Duration (Reward): 99
Episode 190/300 - Duration (Reward): 128
Episode 200/300 - Duration (Reward): 186
Episode 210/300 - Duration (Reward): 82
Episode 220/300 - Duration (Reward): 201
Episode 230/300 - Duration (Reward): 100
Episode 240/300 - Duration (Reward): 78
Episode 250/300 - Duration (Reward): 77
Episode 260/300 - Duration (Reward): 96
Episode 270/300 - Duration (Reward): 102
Episode 280/300 - Duration (Reward): 108
Episode 290/300 - Duration (Reward): 104
Episode 300/300 - Duration (Reward): 110
Training Complete!
Plot saved as 'training_results.png'
```

Figure 4 Starting of training

```
Starting Training...
Episode 10/300 - Duration (Reward): 201
Episode 20/300 - Duration (Reward): 117
Episode 30/300 - Duration (Reward): 154
Episode 40/300 - Duration (Reward): 1000
Episode 50/300 - Duration (Reward): 50
Episode 60/300 - Duration (Reward): 56
Episode 70/300 - Duration (Reward): 56
Episode 80/300 - Duration (Reward): 1000
Episode 90/300 - Duration (Reward): 1000
Episode 100/300 - Duration (Reward): 1000
Episode 110/300 - Duration (Reward): 1000
Episode 120/300 - Duration (Reward): 1000
Episode 130/300 - Duration (Reward): 1000
Episode 140/300 - Duration (Reward): 1000
Episode 150/300 - Duration (Reward): 1000
Episode 160/300 - Duration (Reward): 1000
Episode 170/300 - Duration (Reward): 1000
Episode 180/300 - Duration (Reward): 1000
Episode 190/300 - Duration (Reward): 1000
Episode 200/300 - Duration (Reward): 1000
Episode 210/300 - Duration (Reward): 1000
Episode 220/300 - Duration (Reward): 1000
Episode 230/300 - Duration (Reward): 1000
Episode 240/300 - Duration (Reward): 1000
Episode 250/300 - Duration (Reward): 1000
Episode 260/300 - Duration (Reward): 1000
Episode 270/300 - Duration (Reward): 1000
Episode 280/300 - Duration (Reward): 1000
Episode 290/300 - Duration (Reward): 1000
Episode 300/300 - Duration (Reward): 1000
Training Complete!
Plot saved as 'training_results.png'
```

Figure 3 starting of learning

As seen in the plot, the agent initially performs randomly (low reward duration), but as the neural network converges, the episode duration consistently approaches the maximum of 500 steps. The 100-episode moving average demonstrates stable learning and solving of the environment

1. Blue line (Raw performance):

The blue line shows the agent's score in each episode. It is noisy and fluctuates, which is normal in Reinforcement Learning because:

- **Exploration:** The agent sometimes takes random actions (controlled by **epsilon**, ϵ) to discover better strategies.
- **Learning Instability:** As the neural network updates, it may temporarily perform worse before improving again.

2. The red line (100-Episodes moving average):

The red line shows the average score over the last 100 episodes, making it much smoother than the blue line. It is the most important metric for evaluating a Reinforcement Learning agent. If the red line rises over time, it indicates that the agent is successfully learning and improving.

Advanced calculations

➤ Increasing the Maximum Reward Limit

By default, the CartPole-v1 environment implemented in the Gymnasium library restricts the maximum episode length to 500 steps. Consequently, the maximum achievement is capped at 500. To observe the agent balancing the pole for extended durations (e.g., 1000 or 2000 steps), the environment initialization must be modified.

Below is the necessary adjustment to the code to increase the limit to 2000 steps



```
train_cartpole.py
# Original Initialization (Capped at 500)
env = gym.make("CartPole-v1")

# Modified Initialization (Capped at 1000)
env = gym.make("CartPole-v1", max_episode_steps=1000)
```

Figure 8 Line # 15

➤ Managing Model Memory and Resuming Training

The training script is designed with a state-retention mechanism. Upon starting, it searches the directory for a previously saved model weights file named cartpole_dqn.pth. If this file is detected, the script loads the pre-existing weights to resume training, preventing the loss of prior learning.

The logic for this behavior is highlighted in the code block below:

```

train_cartpole.py

MODEL_PATH = "cartpole_dqn.pth"
if os.path.exists(MODEL_PATH):
    print(f"Loading existing model from '{MODEL_PATH}' to resume training...")
    policy_net.load_state_dict(
        torch.load(MODEL_PATH, map_location=device, weights_only=True)
    )
    EPS_START = 0.1
else:
    print("No existing model found. Starting training from scratch.")

```

Figure 9 Line number 72

➤ Deleting Previous Knowledge

If the objective is to force the neural network to completely "forget" its previous knowledge and learn the environment dynamics from absolute zero, you must intervene manually.

To reset the training:

1. Navigate to your project directory.
2. Locate the file named cartpole_dqn.pth.
3. Delete the file (or rename it if you wish to preserve it as a backup).

Upon the next execution, the script will default to the "Starting training from scratch" condition, re-initializing all neural network weights to random values.

Adding variation in gymnasium environment

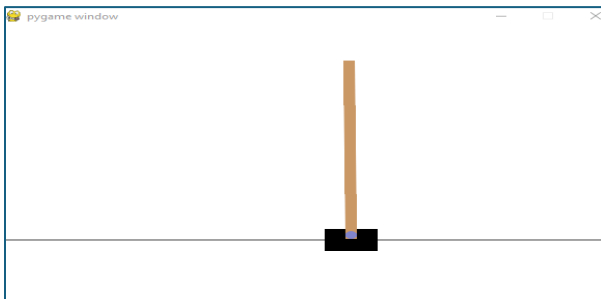
```

train_cartpole.py

env = gym.make("CartPole-v1", max_episode_steps=1000)
env.unwrapped.force_mag = 20.0 # default 10.0
env.unwrapped.gravity = 15.0 # Default is 9.8
env.unwrapped.masspole = 0.5 # Default is 0.1
env.unwrapped.length = 1.0 # Default is 0.5

```

Figure 10 Variations

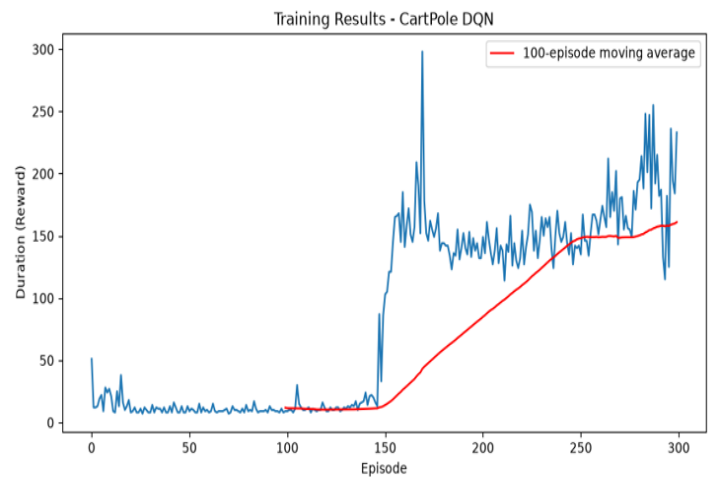


❖ 1st training

```
Starting Training...
Episode 10/300 - Duration (Reward): 27
Episode 20/300 - Duration (Reward): 18
Episode 30/300 - Duration (Reward): 8
Episode 40/300 - Duration (Reward): 8
Episode 50/300 - Duration (Reward): 13
Episode 60/300 - Duration (Reward): 10
Episode 70/300 - Duration (Reward): 11
Episode 80/300 - Duration (Reward): 14
Episode 90/300 - Duration (Reward): 10
Episode 100/300 - Duration (Reward): 9
Episode 110/300 - Duration (Reward): 10
Episode 120/300 - Duration (Reward): 12
Episode 130/300 - Duration (Reward): 12
Episode 140/300 - Duration (Reward): 17
Episode 150/300 - Duration (Reward): 86
Episode 160/300 - Duration (Reward): 185
Episode 170/300 - Duration (Reward): 298
Episode 180/300 - Duration (Reward): 144
Episode 190/300 - Duration (Reward): 140
Episode 200/300 - Duration (Reward): 132
Episode 210/300 - Duration (Reward): 142
Episode 220/300 - Duration (Reward): 132
Episode 230/300 - Duration (Reward): 146
Episode 240/300 - Duration (Reward): 152
Episode 250/300 - Duration (Reward): 142
Episode 260/300 - Duration (Reward): 151
Episode 270/300 - Duration (Reward): 143
Episode 280/300 - Duration (Reward): 193
Episode 290/300 - Duration (Reward): 215
Episode 300/300 - Duration (Reward): 233
Training Complete!
Plot saved as 'training_results.png'
```

```
Starting Evaluation...
Episode 1 finished with a reward of 138.0
Episode 2 finished with a reward of 162.0
Episode 3 finished with a reward of 134.0
Episode 4 finished with a reward of 117.0
Episode 5 finished with a reward of 115.0
Episode 6 finished with a reward of 168.0
Episode 7 finished with a reward of 151.0
Episode 8 finished with a reward of 162.0
Episode 9 finished with a reward of 115.0
Episode 10 finished with a reward of 111.0
Episode 11 finished with a reward of 133.0
Episode 12 finished with a reward of 207.0
Episode 13 finished with a reward of 78.0
Episode 14 finished with a reward of 97.0
Episode 15 finished with a reward of 113.0

D:\assignment>
```



❖ 2nd training

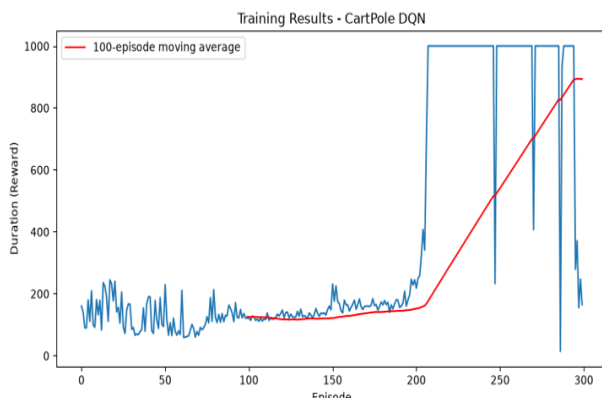


Figure 11 Graphical representation

```
Starting Training...
Episode 10/300 - Duration (Reward): 181
Episode 20/300 - Duration (Reward): 177
Episode 30/300 - Duration (Reward): 166
Episode 40/300 - Duration (Reward): 168
Episode 50/300 - Duration (Reward): 94
Episode 60/300 - Duration (Reward): 68
Episode 70/300 - Duration (Reward): 78
Episode 80/300 - Duration (Reward): 213
Episode 90/300 - Duration (Reward): 153
Episode 100/300 - Duration (Reward): 121
Episode 110/300 - Duration (Reward): 118
Episode 120/300 - Duration (Reward): 136
Episode 130/300 - Duration (Reward): 143
Episode 140/300 - Duration (Reward): 132
Episode 150/300 - Duration (Reward): 148
Episode 160/300 - Duration (Reward): 164
Episode 170/300 - Duration (Reward): 159
Episode 180/300 - Duration (Reward): 173
Episode 190/300 - Duration (Reward): 179
Episode 200/300 - Duration (Reward): 245
Episode 210/300 - Duration (Reward): 1000
Episode 220/300 - Duration (Reward): 1000
Episode 230/300 - Duration (Reward): 1000
Episode 240/300 - Duration (Reward): 1000
Episode 250/300 - Duration (Reward): 1000
Episode 260/300 - Duration (Reward): 1000
Episode 270/300 - Duration (Reward): 1000
Episode 280/300 - Duration (Reward): 1000
Episode 290/300 - Duration (Reward): 1000
Episode 300/300 - Duration (Reward): 165
Training Complete!
```

```
D:\assignment>python test_cartpole.py
Using device: cpu
Loading existing model from 'cartpole_dqn.pt'
C:\Users\Asus\AppData\Local\Programs\Python\Python39\python.exe: UserWarning: pkg_resources is deprecated as an API. See https://www.python.org/dev/peps/pep-0593/ for more details.
ta.py:25: UserWarning: pkg_resources is deprecated as an API. See https://www.python.org/dev/peps/pep-0593/ for more details.
.al as early as 2025-11-30. Refrain from using setuptools to manage your environment.
from pkg_resources import resource_stream,
Starting Evaluation...
Episode 1 finished with a reward of 97.0
Episode 2 finished with a reward of 29.0
Episode 3 finished with a reward of 62.0
Episode 4 finished with a reward of 57.0
Episode 5 finished with a reward of 101.0
Episode 6 finished with a reward of 103.0
Episode 7 finished with a reward of 59.0
Episode 8 finished with a reward of 153.0
Episode 9 finished with a reward of 57.0
Episode 10 finished with a reward of 170.0
Episode 11 finished with a reward of 76.0
Episode 12 finished with a reward of 111.0
Episode 13 finished with a reward of 66.0
Episode 14 finished with a reward of 145.0
Episode 15 finished with a reward of 51.0

D:\assignment>
```


❖ **3^r ---- 6th training:**

```
loading existing model from 'cartpole_dqn.pth' to resume tra
ining...
Starting Training...
Episode 10/300 - Duration (Reward): 1000
Episode 20/300 - Duration (Reward): 70
Episode 30/300 - Duration (Reward): 1000
Episode 40/300 - Duration (Reward): 182
Episode 50/300 - Duration (Reward): 178
Episode 60/300 - Duration (Reward): 172
Episode 70/300 - Duration (Reward): 173
Episode 80/300 - Duration (Reward): 174
Episode 90/300 - Duration (Reward): 178
Episode 100/300 - Duration (Reward): 185
Episode 110/300 - Duration (Reward): 181
Episode 120/300 - Duration (Reward): 182
Episode 130/300 - Duration (Reward): 199
Episode 140/300 - Duration (Reward): 1000
Episode 150/300 - Duration (Reward): 1000
Episode 160/300 - Duration (Reward): 1000
Episode 170/300 - Duration (Reward): 1000
Episode 180/300 - Duration (Reward): 1000
Episode 190/300 - Duration (Reward): 1000
Episode 200/300 - Duration (Reward): 1000
Episode 210/300 - Duration (Reward): 1000
Episode 220/300 - Duration (Reward): 1000
Episode 230/300 - Duration (Reward): 1000
Episode 240/300 - Duration (Reward): 569
Episode 250/300 - Duration (Reward): 1000
Episode 260/300 - Duration (Reward): 1000
Episode 270/300 - Duration (Reward): 1000
Episode 280/300 - Duration (Reward): 1000
Episode 290/300 - Duration (Reward): 1000
Episode 300/300 - Duration (Reward): 1000
Training Complete!
Plot saved as 'training_results.png'
```

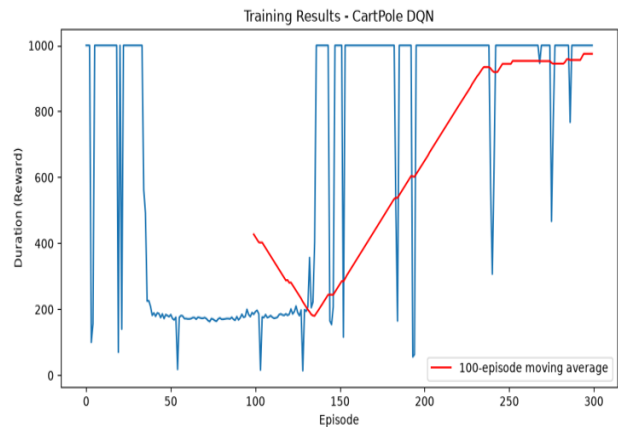


Figure 12 graph

Figure 13 testing and training

❖ 7th Training

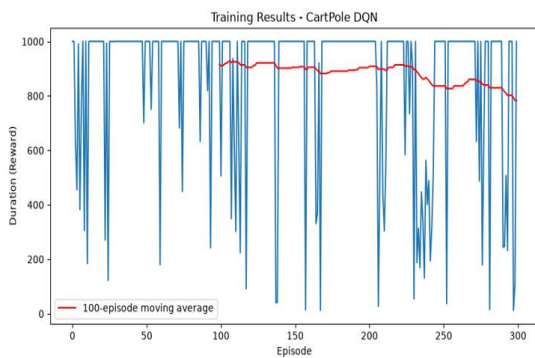


Figure 14 graph

```
Starting Training...
Episode 10/300 - Duration (Reward): 1000
Episode 20/300 - Duration (Reward): 1000
Episode 30/300 - Duration (Reward): 1000
Episode 40/300 - Duration (Reward): 1000
Episode 50/300 - Duration (Reward): 1000
Episode 60/300 - Duration (Reward): 180
Episode 70/300 - Duration (Reward): 1000
Episode 80/300 - Duration (Reward): 1000
Episode 90/300 - Duration (Reward): 1000
Episode 100/300 - Duration (Reward): 1000
Episode 110/300 - Duration (Reward): 602
Episode 120/300 - Duration (Reward): 1000
Episode 130/300 - Duration (Reward): 1000
Episode 140/300 - Duration (Reward): 1000
Episode 150/300 - Duration (Reward): 1000
Episode 160/300 - Duration (Reward): 1000
Episode 170/300 - Duration (Reward): 1000
Episode 180/300 - Duration (Reward): 1000
Episode 190/300 - Duration (Reward): 1000
Episode 200/300 - Duration (Reward): 1000
Episode 210/300 - Duration (Reward): 440
Episode 220/300 - Duration (Reward): 1000
Episode 230/300 - Duration (Reward): 914
Episode 240/300 - Duration (Reward): 401
Episode 250/300 - Duration (Reward): 1000
Episode 260/300 - Duration (Reward): 1000
Episode 270/300 - Duration (Reward): 1000
Episode 280/300 - Duration (Reward): 1000
Episode 290/300 - Duration (Reward): 1000
Episode 300/300 - Duration (Reward): 1000
Training Complete!

from pkg_resources import resource_stream
Starting Evaluation...
Episode 1 finished with a reward of 1000.0
Episode 2 finished with a reward of 29.0
Episode 3 finished with a reward of 60.0
Episode 4 finished with a reward of 93.0
Episode 5 finished with a reward of 1000.0
Episode 6 finished with a reward of 27.0
Episode 7 finished with a reward of 69.0
Episode 8 finished with a reward of 136.0
Episode 9 finished with a reward of 121.0
Episode 10 finished with a reward of 29.0
Episode 11 finished with a reward of 88.0
Episode 12 finished with a reward of 1000.0
Episode 13 finished with a reward of 144.0
Episode 14 finished with a reward of 196.0
Episode 15 finished with a reward of 318.0

D:\assignment\
```

❖ 8th training

]

```
Starting Training...
Episode 10/300 - Duration (Reward): 1000
Episode 20/300 - Duration (Reward): 1000
Episode 30/300 - Duration (Reward): 1000
Episode 40/300 - Duration (Reward): 1000
Episode 50/300 - Duration (Reward): 1000
Episode 60/300 - Duration (Reward): 1000
Episode 70/300 - Duration (Reward): 1000
Episode 80/300 - Duration (Reward): 1000
Episode 90/300 - Duration (Reward): 1000
Episode 100/300 - Duration (Reward): 1000
Episode 110/300 - Duration (Reward): 1000
Episode 120/300 - Duration (Reward): 1000
Episode 130/300 - Duration (Reward): 659
Episode 140/300 - Duration (Reward): 1000
Episode 150/300 - Duration (Reward): 1000
Episode 160/300 - Duration (Reward): 1000
Episode 170/300 - Duration (Reward): 289
Episode 180/300 - Duration (Reward): 1000
Episode 190/300 - Duration (Reward): 1000
Episode 200/300 - Duration (Reward): 209
Episode 210/300 - Duration (Reward): 203
Episode 220/300 - Duration (Reward): 725
Episode 230/300 - Duration (Reward): 1000
Episode 240/300 - Duration (Reward): 1000
Episode 250/300 - Duration (Reward): 503
Episode 260/300 - Duration (Reward): 1000
Episode 270/300 - Duration (Reward): 1000
Episode 280/300 - Duration (Reward): 1000
Episode 290/300 - Duration (Reward): 1000
Episode 300/300 - Duration (Reward): 1000
Training Complete!

Using device: cpu
Loading existing model from 'cartpole_dqn.pt'
Warning: pkg_resources is deprecated as an API.
See https://setuptools.pypa.io/en/latest/pkg_resources.html.
The pkg_resources package is slated for removal as early as 2025-07-06.
Please upgrade to setuptools 68.1 or later to remove this package or pin to Setuptools<81.
from pkg_resources import resource_stream,
Starting Evaluation...
Episode 1 finished with a reward of 1000.0
Episode 2 finished with a reward of 1000.0
Episode 3 finished with a reward of 1000.0
Episode 4 finished with a reward of 1000.0
Episode 5 finished with a reward of 1000.0
Episode 6 finished with a reward of 51.0
Episode 7 finished with a reward of 1000.0
Episode 8 finished with a reward of 67.0
Episode 9 finished with a reward of 32.0
Episode 10 finished with a reward of 1000.0
Episode 11 finished with a reward of 1000.0
Episode 12 finished with a reward of 1000.0
Episode 13 finished with a reward of 1000.0
Episode 14 finished with a reward of 1000.0
Episode 15 finished with a reward of 1000.0

D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
```

Figure 16 Training and Testing

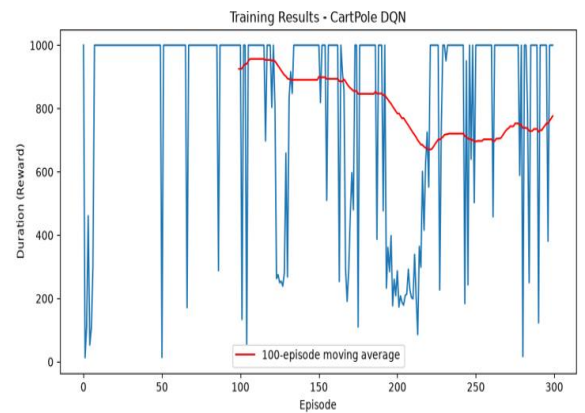


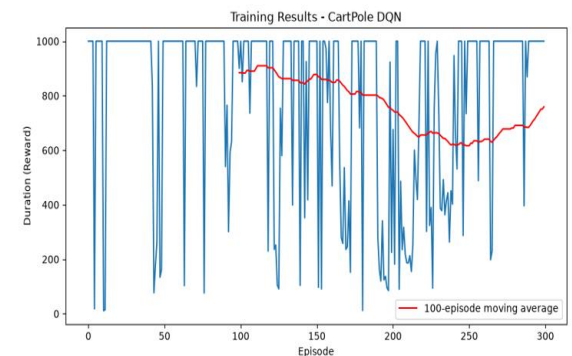
Figure 15 Graph

❖ 9th training

```
Using device: cpu
Loading existing model from 'cartpole_dqn.pt'
Starting Training...
Episode 10/300 - Duration (Reward): 1000
Episode 20/300 - Duration (Reward): 1000
Episode 30/300 - Duration (Reward): 1000
Episode 40/300 - Duration (Reward): 1000
Episode 50/300 - Duration (Reward): 1000
Episode 60/300 - Duration (Reward): 1000
Episode 70/300 - Duration (Reward): 1000
Episode 80/300 - Duration (Reward): 1000
Episode 90/300 - Duration (Reward): 1000
Episode 100/300 - Duration (Reward): 900
Episode 110/300 - Duration (Reward): 1000
Episode 120/300 - Duration (Reward): 1000
Episode 130/300 - Duration (Reward): 1000
Episode 140/300 - Duration (Reward): 104
Episode 150/300 - Duration (Reward): 1000
Episode 160/300 - Duration (Reward): 676
Episode 170/300 - Duration (Reward): 236
Episode 180/300 - Duration (Reward): 1000
Episode 190/300 - Duration (Reward): 1000
Episode 200/300 - Duration (Reward): 225
Episode 210/300 - Duration (Reward): 187
Episode 220/300 - Duration (Reward): 1000
Episode 230/300 - Duration (Reward): 1000
Episode 240/300 - Duration (Reward): 402
Episode 250/300 - Duration (Reward): 1000
Episode 260/300 - Duration (Reward): 1000
Episode 270/300 - Duration (Reward): 1000
Episode 280/300 - Duration (Reward): 1000
Episode 290/300 - Duration (Reward): 1000
Episode 300/300 - Duration (Reward): 1000
Training Complete!

pygame/pkgdata.py:25: UserWarning: pkg_resources
is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html.
The pkg_resources package is slated for removal as early as 2025-07-06.
Please upgrade to setuptools 68.1 or later to remove this package or pin to Setuptools<81.
from pkg_resources import resource_stream,
Starting Evaluation...
Episode 1 finished with a reward of 1000.0
Episode 2 finished with a reward of 1000.0
Episode 3 finished with a reward of 1000.0
Episode 4 finished with a reward of 1000.0
Episode 5 finished with a reward of 1000.0
Episode 6 finished with a reward of 1000.0
Episode 7 finished with a reward of 1000.0
Episode 8 finished with a reward of 1000.0
Episode 9 finished with a reward of 1000.0
Episode 10 finished with a reward of 1000.0
Episode 11 finished with a reward of 1000.0
Episode 12 finished with a reward of 1000.0
Episode 13 finished with a reward of 1000.0
Episode 14 finished with a reward of 1000.0
Episode 15 finished with a reward of 1000.0

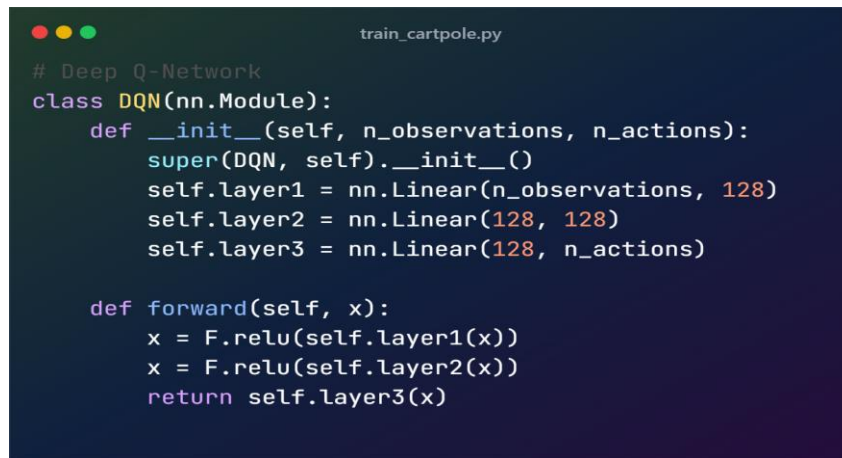
D:\assignment>
D:\assignment>
D:\assignment>
```



Now the model is giving better outcomes which means it is now well trained

Adding variations in model

- **change the neural network and activation function:**
Currently it has 2 neural networks of 128 nodes each.



```
train_cartpole.py

# Deep Q-Network
class DQN(nn.Module):
    def __init__(self, n_observations, n_actions):
        super(DQN, self).__init__()
        self.layer1 = nn.Linear(n_observations, 128)
        self.layer2 = nn.Linear(128, 128)
        self.layer3 = nn.Linear(128, n_actions)

    def forward(self, x):
        x = F.relu(self.layer1(x))
        x = F.relu(self.layer2(x))
        return self.layer3(x)
```

Figure 17 Before changes

Changed it from 128 to 256 nodes (a bigger brain) and added a self.layer4 to make the network deeper
And changed the activation function from **f.relu** to **torch.sigmoid**.



```
train_cartpole.py

# Deep Q-Network
class DQN(nn.Module):
    def __init__(self, n_observations, n_actions):
        super(DQN, self).__init__()
        self.layer1 = nn.Linear(n_observations, 256)
        self.layer2 = nn.Linear(256, 256)
        self.layer3 = nn.Linear(256, 256)
        self.layer4 = nn.Linear(256, n_actions)

    def forward(self, x):
        x = torch.sigmoid(self.layer1(x))
        x = torch.sigmoid(self.layer2(x))
        x = torch.sigmoid(self.layer3(x))
        return self.layer4(x)
```

Figure 18 After Changes

- **Changes hyperparameter (the learning rate):**
 - **LR = 1e-4 (Learning Rate):** This controls how big of a step the network takes when updating its knowledge.
 - **BATCH_SIZE = 128:** How many memories it looks at simultaneously while training.


```
train_cartpole.py

BATCH_SIZE = 128
GAMMA = 0.99
EPS_START = 0.9
EPS_END = 0.05
EPS_DECAY = 1000
TAU = 0.005
LR = 1e-4
```

Figure 19 Before changes

- **LR = 1e-4 (Learning Rate):** Changes it to 1e-3 (learns faster, but might be unstable)
- **BATCH_SIZE =** Raising it to 256.

```
train_cartpole.py

BATCH_SIZE = 256
GAMMA = 0.99
EPS_START = 0.9
EPS_END = 0.05
EPS_DECAY = 1000
TAU = 0.005 # Target network update rate
LR = 1e-3 # Learning rate
```

Figure 20 After changes

➤ **CHANGE THE EXPLORATION RATE (ϵ -GREEDY):**

EPS_DECAY = 1000. If we lower this to 500, the agent stops acting randomly much earlier in the training. If you raise it to 2000, it will continue to explore randomly for a much longer time.

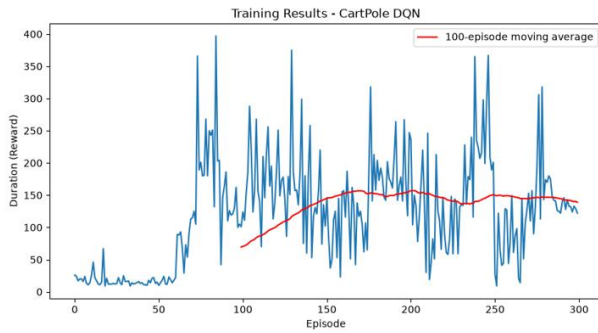
```
train_cartpole.py

BATCH_SIZE = 256
GAMMA = 0.99
EPS_START = 0.9
EPS_END = 0.05
EPS_DECAY = 2000
TAU = 0.005 # Target network update rate
LR = 1e-3 # Learning rate
```

Changing it to 2000 so that model will explore more

Start training the model after variations:

❖ 1st training



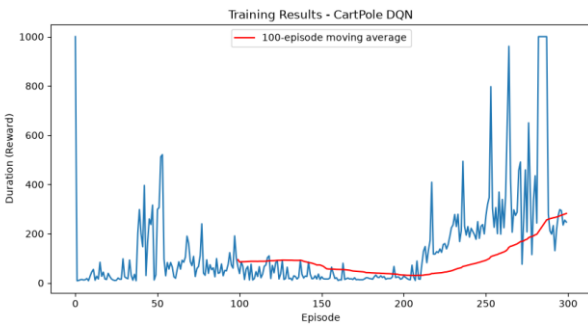
```
No existing model found. Starting training...
Starting Training...
Episode 10/300 - Duration (Reward): 14
Episode 20/300 - Duration (Reward): 21
Episode 30/300 - Duration (Reward): 25
Episode 40/300 - Duration (Reward): 13
Episode 50/300 - Duration (Reward): 15
Episode 60/300 - Duration (Reward): 18
Episode 70/300 - Duration (Reward): 11
Episode 80/300 - Duration (Reward): 18
Episode 90/300 - Duration (Reward): 16
Episode 100/300 - Duration (Reward): 1
Episode 110/300 - Duration (Reward): 1
Episode 120/300 - Duration (Reward): 1
Episode 130/300 - Duration (Reward): 3
Episode 140/300 - Duration (Reward): 1
Episode 150/300 - Duration (Reward): 1
Episode 160/300 - Duration (Reward): 1
Episode 170/300 - Duration (Reward): 1
Episode 180/300 - Duration (Reward): 1
Episode 190/300 - Duration (Reward): 1
Episode 200/300 - Duration (Reward): 2
Episode 210/300 - Duration (Reward): 3
Episode 220/300 - Duration (Reward): 1
Episode 230/300 - Duration (Reward): 1
Episode 240/300 - Duration (Reward): 2
Episode 250/300 - Duration (Reward): 2
Episode 260/300 - Duration (Reward): 7
Episode 270/300 - Duration (Reward): 1
Episode 280/300 - Duration (Reward): 1
Episode 290/300 - Duration (Reward): 1
Episode 300/300 - Duration (Reward): 1
Training Complete!
```

```
1. The pkg_resources package is slated for
11-30. Refrain from using this package or
from pkg_resources import resource_stream
Starting Evaluation...
Episode 1 finished with a reward of 56.0
Episode 2 finished with a reward of 54.0
Episode 3 finished with a reward of 59.0
Episode 4 finished with a reward of 63.0
Episode 5 finished with a reward of 55.0
Episode 6 finished with a reward of 60.0
Episode 7 finished with a reward of 63.0
Episode 8 finished with a reward of 53.0
Episode 9 finished with a reward of 63.0
Episode 10 finished with a reward of 57.0
Episode 11 finished with a reward of 61.0
Episode 12 finished with a reward of 59.0
Episode 13 finished with a reward of 59.0
Episode 14 finished with a reward of 65.0
Episode 15 finished with a reward of 52.0
```

```
D:\assignment>python --version
Python 3.13.1
```

```
D:\assignment>
```

❖ 2nd training

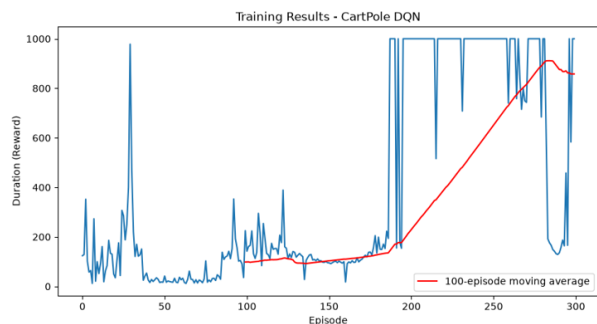


```
Starting Training...
Episode 10/300 - Duration (Reward): 28
Episode 20/300 - Duration (Reward): 14
Episode 30/300 - Duration (Reward): 98
Episode 40/300 - Duration (Reward): 298
Episode 50/300 - Duration (Reward): 33
Episode 60/300 - Duration (Reward): 65
Episode 70/300 - Duration (Reward): 160
Episode 80/300 - Duration (Reward): 31
Episode 90/300 - Duration (Reward): 77
Episode 100/300 - Duration (Reward): 65
Episode 110/300 - Duration (Reward): 18
Episode 120/300 - Duration (Reward): 18
Episode 130/300 - Duration (Reward): 64
Episode 140/300 - Duration (Reward): 20
Episode 150/300 - Duration (Reward): 14
Episode 160/300 - Duration (Reward): 21
Episode 170/300 - Duration (Reward): 14
Episode 180/300 - Duration (Reward): 18
Episode 190/300 - Duration (Reward): 15
Episode 200/300 - Duration (Reward): 18
Episode 210/300 - Duration (Reward): 16
Episode 220/300 - Duration (Reward): 117
Episode 230/300 - Duration (Reward): 224
Episode 240/300 - Duration (Reward): 229
Episode 250/300 - Duration (Reward): 199
Episode 260/300 - Duration (Reward): 198
Episode 270/300 - Duration (Reward): 287
Episode 280/300 - Duration (Reward): 337
Episode 290/300 - Duration (Reward): 211
Episode 300/300 - Duration (Reward): 247
Training Complete!
```

```
D:\assignment>python test_cartpole.py
Using device: cpu
Loading existing model from 'cartpole_dqn.
C:\Users\Asus\AppData\Local\Programs\Python
y:25: UserWarning: pkg_resources is deprec
/en/latest/pkg_resources.html. The pkg_res
as 2025-11-30. Refrain from using this pac
from pkg_resources import resource_stream
Starting Evaluation...
Episode 1 finished with a reward of 55.0
Episode 2 finished with a reward of 63.0
Episode 3 finished with a reward of 54.0
Episode 4 finished with a reward of 65.0
Episode 5 finished with a reward of 71.0
Episode 6 finished with a reward of 57.0
Episode 7 finished with a reward of 59.0
Episode 8 finished with a reward of 56.0
Episode 9 finished with a reward of 66.0
Episode 10 finished with a reward of 57.0
Episode 11 finished with a reward of 56.0
Episode 12 finished with a reward of 52.0
Episode 13 finished with a reward of 66.0
Episode 14 finished with a reward of 53.0
Episode 15 finished with a reward of 59.0
```

```
D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
D:\assignment>
```

❖ 3rd training:



```
Starting Training...
Episode 10/300 - Duration (Reward): 100
Episode 20/300 - Duration (Reward): 50
Episode 30/300 - Duration (Reward): 978
Episode 40/300 - Duration (Reward): 54
Episode 50/300 - Duration (Reward): 17
Episode 60/300 - Duration (Reward): 37
Episode 70/300 - Duration (Reward): 27
Episode 80/300 - Duration (Reward): 32
Episode 90/300 - Duration (Reward): 142
Episode 100/300 - Duration (Reward): 225
Episode 110/300 - Duration (Reward): 84
Episode 120/300 - Duration (Reward): 130
Episode 130/300 - Duration (Reward): 138
Episode 140/300 - Duration (Reward): 129
Episode 150/300 - Duration (Reward): 96
Episode 160/300 - Duration (Reward): 100
Episode 170/300 - Duration (Reward): 106
Episode 180/300 - Duration (Reward): 135
Episode 190/300 - Duration (Reward): 1000
Episode 200/300 - Duration (Reward): 1000
Episode 210/300 - Duration (Reward): 1000
Episode 220/300 - Duration (Reward): 1000
Episode 230/300 - Duration (Reward): 1000
Episode 240/300 - Duration (Reward): 1000
Episode 250/300 - Duration (Reward): 1000
Episode 260/300 - Duration (Reward): 740
Episode 270/300 - Duration (Reward): 749
Episode 280/300 - Duration (Reward): 684
Episode 290/300 - Duration (Reward): 129
Episode 300/300 - Duration (Reward): 1000
Training Complete!
Plot saved as 'training_results.png'

Using device: cpu
Loading existing model from 'cartpole_dqn.pth'
C:\Users\Asus\AppData\Local\Programs\Python\Python38-64\Scripts\python.exe: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal in Python 3.12. Refrain from using this package or pip install setuptools<67.0.0 to avoid this warning.
Starting Evaluation...
Episode 1 finished with a reward of 1000.0
Episode 2 finished with a reward of 1000.0
Episode 3 finished with a reward of 1000.0
Episode 4 finished with a reward of 1000.0
Episode 5 finished with a reward of 770.0
Episode 6 finished with a reward of 1000.0
Episode 7 finished with a reward of 1000.0
Episode 8 finished with a reward of 998.0
Episode 9 finished with a reward of 1000.0
Episode 10 finished with a reward of 429.0
Episode 11 finished with a reward of 1000.0
Episode 12 finished with a reward of 1000.0
Episode 13 finished with a reward of 1000.0
Episode 14 finished with a reward of 1000.0
Episode 15 finished with a reward of 458.0

D:\assignment>1
'1' is not recognized as an internal or external command,
operable program or batch file.

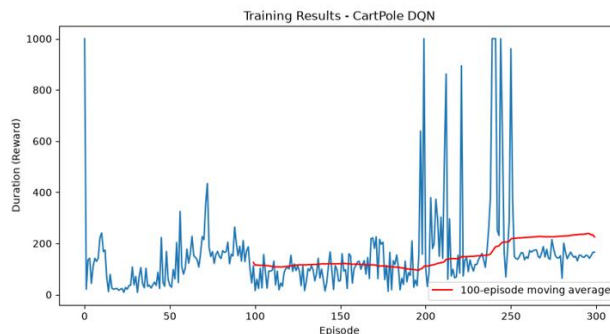
D:\assignment>1
'1' is not recognized as an internal or external command,
operable program or batch file.
```

❖ 4th ---- 10th training:

```
Starting Training...
Episode 10/300 - Duration (Reward): 222
Episode 20/300 - Duration (Reward): 24
Episode 30/300 - Duration (Reward): 38
Episode 40/300 - Duration (Reward): 26
Episode 50/300 - Duration (Reward): 60
Episode 60/300 - Duration (Reward): 103
Episode 70/300 - Duration (Reward): 227
Episode 80/300 - Duration (Reward): 150
Episode 90/300 - Duration (Reward): 199
Episode 100/300 - Duration (Reward): 113
Episode 110/300 - Duration (Reward): 92
Episode 120/300 - Duration (Reward): 106
Episode 130/300 - Duration (Reward): 16
Episode 140/300 - Duration (Reward): 100
Episode 150/300 - Duration (Reward): 53
Episode 160/300 - Duration (Reward): 113
Episode 170/300 - Duration (Reward): 222
Episode 180/300 - Duration (Reward): 165
Episode 190/300 - Duration (Reward): 50
Episode 200/300 - Duration (Reward): 1000
Episode 210/300 - Duration (Reward): 303
Episode 220/300 - Duration (Reward): 154
Episode 230/300 - Duration (Reward): 118
Episode 240/300 - Duration (Reward): 1000
Episode 250/300 - Duration (Reward): 287
Episode 260/300 - Duration (Reward): 148
Episode 270/300 - Duration (Reward): 188
Episode 280/300 - Duration (Reward): 152
Episode 290/300 - Duration (Reward): 132
Episode 300/300 - Duration (Reward): 166
Training Complete!

py:25: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal in Python 3.12. Refrain from using this package or pip install setuptools<67.0.0 to avoid this warning.
Starting Evaluation...
Episode 1 finished with a reward of 1000.0
Episode 2 finished with a reward of 1000.0
Episode 3 finished with a reward of 1000.0
Episode 4 finished with a reward of 1000.0
Episode 5 finished with a reward of 1000.0
Episode 6 finished with a reward of 1000.0
Episode 7 finished with a reward of 1000.0
Episode 8 finished with a reward of 1000.0
Episode 9 finished with a reward of 1000.0
Episode 10 finished with a reward of 1000.0
Episode 11 finished with a reward of 1000.0
Episode 12 finished with a reward of 1000.0
Episode 13 finished with a reward of 1000.0
Episode 14 finished with a reward of 1000.0
Episode 15 finished with a reward of 1000.0

D:\assignment>
```



The model is exploring more in a messy form because of LR = 1-e3

How to Run the model:

1. Install dependencies: `pip install torch gymnasium matplotlib NumPy`
2. To resume training the model: Run → `python train_cartpole.py`
3. To watch the trained agent, play: Run → `python test_cartpole.py` (*Note: The script automatically loads the pre-trained weights from `cartpole_dqn.pth`*)

Conclusion:

This assignment successfully demonstrated the application of Deep Reinforcement Learning to solve the classic CartPole control problem. The initial implementation utilized a Deep Q-Network (DQN) paired with an Experience Replay Buffer and a Target Network, which allowed the agent to stably learn an optimal policy and reach the environment's maximum reward limit.

Beyond the base requirements, several advanced experiments were conducted to test the robustness and flexibility of the DQN algorithm:

1. **Extreme Physics Simulation:** The underlying physical constraints of the Gymnasium environment were heavily modified to simulate extreme conditions. Gravity increased from 9.8 to 50.0, the pole mass increased to 1.0, and the force magnitude applied by the cart quadrupled to 40.0. This created a highly volatile environment where the standard policy immediately failed.
2. **Network and Training Scaling:** To accommodate the drastically increased complexity of the new physics engine, the neural network architecture was deepened to four layers utilizing Sigmoid activations. Furthermore, the environment's maximum reward threshold was raised to 1000 steps, and the training cycle was extended to 900 episodes, giving the deeper network sufficient data and time to converge.
3. **State Retention (Model Saving/Loading):** A persistence mechanism was implemented to save the neural network's learned weights (.pth files). This allowed the agent to pause and resume training over multiple sessions without catastrophic forgetting, significantly improving the efficiency of the training process by lowering the exploration rate (ϵ) upon resuming.